# The Binomial Theorem

ClipSAT Printable Practice Suite (clipsat.org) • Topic 9.03

**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Binomial coefficients**

Evaluate each binomial coefficient:

**a** $\binom{6}{2} = 15$ **b** $\binom{8}{0} = 1$ **c** $\binom{5}{5} = 1$ **d** $\binom{10}{3} = 120$

(a) $= 15$; (b) $= 1$; (c) $= 1$; (d) $= 120$

**2** **[Foundational]** • **Target: Binomial coefficients**

Write down row 4 of Pascal's triangle (the row beginning $1, 4, \ldots$) and explain how it relates to the expansion of $(a + b)^4$.

Row 4 is $1, 4, 6, 4, 1$ — these are the coefficients of $a^4, a^3b, a^2b^2, ab^3, b^4$ in the expansion.

**3** **[Foundational]** • **Target: Expansions and specific terms**

Expand $(x + 2)^4$ fully.

$x^4 + 8x^3 + 24x^2 + 32x + 16$.

**4** **[Foundational]** • **Target: Expansions and specific terms**

Find the coefficient of $x^3$ in the expansion of $(2x - 1)^5$.

Term: $\binom{5}{2}(2x)^3(-1)^2 = 10 \cdot 8x^3 = 80x^3$. Coefficient: $80$.

**5** **[Foundational]** • **Target: Expansions and specific terms**

Find the term containing $y^4$ in the expansion of $(x + y)^7$.

$\binom{7}{4}x^3y^4 = 35x^3y^4$.

**6** **[Foundational]** • **Target: Expansions and specific terms**

Use the binomial theorem to expand $(1 + 0.02)^5$ to the $x^2$-style second-order term, and hence estimate $1.02^5$ to four decimal places.

$1 + 5(0.02) + 10(0.02)^2 + \cdots = 1 + 0.1 + 0.004 + \cdots \approx 1.1041$ (true value $1.10408\ldots$).

**7** **[Foundational]** • **Target: More expansions and terms**

Expand $(2x - y)^3$ fully.

$8x^3 - 12x^2y + 6xy^2 - y^3$.

© ClipSAT — clipsat.org | All Rights Reserved

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** **[Intermediate]** • **Target: More expansions and terms**

Find the middle term of the expansion of $(x+3)^6$ (the term with $x^3$).

Middle term is the 4th term: $\binom{6}{3}x^3(3)^3 = 20 \cdot 27x^3 = 540x^3$.

**9** **[Intermediate]** • **Target: More expansions and terms**

Find the constant term in the expansion of $\left(x^2 + \frac{1}{x}\right)^6$.

General term: $\binom{6}{k}(x^2)^{6-k}\left(\frac{1}{x}\right)^k = \binom{6}{k}x^{12-3k}$. Constant term needs $12 - 3k = 0$, so $k = 4$: $\binom{6}{4} = 15$.

**10** **[Intermediate]** • **Target: Sums of binomial coefficients**

Evaluate $\binom{5}{0} + \binom{5}{1} + \binom{5}{2} + \binom{5}{3} + \binom{5}{4} + \binom{5}{5}$ without computing each term, using the identity $\sum_{k=0}^{n}\binom{n}{k} = 2^n$.

$= 2^5 = 32$.

**11** **[Intermediate]** • **Target: Sums of binomial coefficients**

Use the fact that $\binom{n}{k} = \binom{n}{n-k}$ (symmetry of Pascal's triangle) to explain why $\binom{9}{2} = \binom{9}{7}$ without computing either value directly.

Choosing $2$ items to include from $9$ is equivalent to choosing $7$ items to exclude, so the two counts must be equal — this is the symmetry property $\binom{n}{k} = \binom{n}{n-k}$ with $n = 9$.

**12** **[Intermediate]** • **Target: Further expansions**

Find the coefficient of $x^4y^3$ in the expansion of $(x+2y)^7$.

Term: $\binom{7}{3}x^4(2y)^3 = 35 \cdot 8\,x^4y^3 = 280x^4y^3$. Coefficient: $280$.

**13** **[Intermediate]** • **Target: Further expansions**

Find the first three terms in the expansion of $(1-2x)^6$ in ascending powers of $x$.

$\binom{6}{0}1^6 + \binom{6}{1}1^5(-2x) + \binom{6}{2}1^4(-2x)^2 + \cdots = 1 - 12x + 60x^2 - \cdots$.

**14** **[Intermediate]** • **Target: Further expansions**

Use the binomial theorem with $(1-0.01)^4$ to approximate $0.99^4$ to four decimal places, using the first three terms.

$1 + 4(-0.01) + 6(-0.01)^2 + \cdots = 1 - 0.04 + 0.0006 = 0.9606$ (true value $0.96059601$).

---

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

© ClipSAT — clipsat.org | All Rights Reserved

**15** [Challenging] • Target: Deriving Pascal's triangle recursively • Cross-Lesson: 5.01 Vectors: Components and Operations

State Pascal's Rule, $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$, and use it to compute $\binom{6}{3}$ from row 5 of Pascal's triangle ($1, 5, 10, 10, 5, 1$).

$\binom{6}{3} = \binom{5}{2} + \binom{5}{3} = 10 + 10 = 20$.

**16** [Challenging] • Target: The general term formula • Cross-Lesson: 5.02 The Dot Product

State the general term (the $(k+1)$th term) formula for the expansion of $(a+b)^n$, and use it to find the 5th term of $(x-2)^8$ directly (without expanding the whole thing).

General term: $T_{k+1} = \binom{n}{k} a^{n-k} b^k$. For the 5th term, $k = 4$: $T_5 = \binom{8}{4} x^4 (-2)^4 = 70 \cdot x^4 \cdot 16 = 1120x^4$.

**17** [Challenging] • Target: Combinatorial interpretation of binomial coefficients • Cross-Lesson: 5.03 Parametric Equations

Explain the combinatorial meaning of $\binom{n}{k}$, and use it to explain why $\binom{n}{0} = \binom{n}{n} = 1$ for any $n$.

$\binom{n}{k}$ counts the number of ways to choose $k$ items from a set of $n$ (order doesn't matter). $\binom{n}{0} = 1$ because there is exactly one way to choose nothing (the empty selection), and $\binom{n}{n} = 1$ because there is exactly one way to choose everything (select the whole set).

**18** [Challenging] • Target: Applying the binomial theorem to probability-style counting • Cross-Lesson: 6.01 Polar Coordinates and Conversions

A fair coin is flipped 5 times. Using $\binom{5}{k}$, find the number of outcomes with exactly 3 heads, and state the total number of possible outcomes (using $2^5$).

Outcomes with exactly 3 heads: $\binom{5}{3} = 10$. Total possible outcomes: $2^5 = 32$.

**19** [Challenging] • Target: A closing challenge problem • Cross-Lesson: 6.02 Polar Equations and Graphs

Find the sum of all the coefficients in the expansion of $(3x-2)^4$, using the fact that setting $x = 1$ gives the sum of all coefficients.

Substituting $x = 1$: $(3(1) - 2)^4 = (1)^4 = 1$. The sum of all coefficients in the expansion is $1$.

**20** [Challenging] • Target: A closing challenge problem • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

Find the coefficient of $x^5$ in the expansion of $(x^2+3)(x-1)^6$, by finding the relevant terms from $(x-1)^6$ first.

From $(x-1)^6$, the terms of degree $3$ and $5$ are needed (since multiplying by $x^2$ or by the constant $3$ must produce $x^5$). Term with $x^3$: $\binom{6}{3} x^3 (-1)^3 = -20x^3$; multiplying by $x^2$ gives $-20x^5$. Term with $x^5$: $\binom{6}{1} x^5 (-1)^1 = -6x^5$; multiplying by $3$ gives $-18x^5$. Total coefficient: $-20 + (-18) = -38$.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

**21** [Foundational] • Target: Pascal's Triangle and Combinations

State the formula for the combination $\binom{n}{r}$.

**a.** $\frac{n!}{r!\,(n-r)!}$ ✓

**b.** $\frac{n!}{(n-r)!}$

**c.** $\frac{n!}{r!}$

**d.** $n \times r$

$\binom{n}{r} = \frac{n!}{r!\,(n-r)!}$ gives the coefficient of the $(r+1)$-th term in $(a+b)^n$.

**22** [Foundational] • Target: Pascal's Triangle and Combinations

Evaluate $\binom{6}{2}$.

**a.** 15 ✓

**b.** 30

**c.** 12

**d.** 20

$\binom{6}{2} = \frac{6 \times 5}{2 \times 1} = 15$.

**23** [Foundational] • Target: Pascal's Triangle and Combinations

What is row 4 of Pascal's Triangle (corresponding to $n = 4$)?

**a.** 1, 4, 6, 4, 1 ✓

**b.** 1, 3, 3, 1

**c.** 1, 5, 10, 10, 5, 1

**d.** 1, 4, 8, 4, 1

Row 4 is $\binom{4}{0} = 1$, $\binom{4}{1} = 4$, $\binom{4}{2} = 6$, $\binom{4}{3} = 4$, $\binom{4}{4} = 1$.

**24** [Foundational] • Target: Expanding Binomials

State the Binomial Theorem expansion for $(a+b)^n$.

**a.** $\sum_{r=0}^{n} \binom{n}{r} a^{n-r} b^r$ ✓

**b.** $\sum_{r=0}^{n} a^{n-r} b^r$

**c.** $\sum_{r=1}^{n} \binom{n}{r} a^r b^{n-r}$

**d.** $\sum_{r=0}^{n} n!\, a^r b^r$

The Binomial Theorem states $(a+b)^n = \sum_{r=0}^{n} \binom{n}{r} a^{n-r} b^r$.

© ClipSAT — clipsat.org | All Rights Reserved

**25** [Foundational] • Target: Expanding Binomials

Expand $(x+2)^4$.

**a.** $x^4+8x^3+24x^2+32x+16$ ✓

**b.** $x^4+4x^3+6x^2+4x+16$

**c.** $x^4+16$

**d.** $x^4+8x^3+16x^2+32x+16$

$\binom{4}{0}x^4+\binom{4}{1}x^3(2)+\binom{4}{2}x^2(4)+\binom{4}{3}x(8)+\binom{4}{4}(16)=x^4+8x^3+24x^2+32x+16$.

**26** [Foundational] • Target: Expanding Binomials

Find the first three terms in the expansion of $(2x-1)^5$ in descending powers of $x$.

**a.** $32x^5-80x^4+80x^3$ ✓

**b.** $32x^5-16x^4+8x^3$

**c.** $32x^5+80x^4+80x^3$

**d.** $32x^5-40x^4+20x^3$

Term 1: $\binom{5}{0}(2x)^5=32x^5$. Term 2: $\binom{5}{1}(2x)^4(-1)=5(16x^4)(-1)=-80x^4$. Term 3: $\binom{5}{2}(2x)^3(-1)^2=10(8x^3)(1)=80x^3$.

**27** [Foundational] • Target: Finding Specific Terms

State the formula for the $(r+1)$-th term in the expansion of $(a+b)^n$.

**a.** $T_{r+1}=\binom{n}{r}a^{n-r}b^r$ ✓

**b.** $T_{r+1}=\binom{n}{r+1}a^{n-r}b^r$

**c.** $T_{r+1}=\binom{n}{r}a^rb^{n-r}$

**d.** $T_{r+1}=a^{n-r}b^r$

The general term is given by $T_{r+1}=\binom{n}{r}a^{n-r}b^r$ where $r$ indexes from 0 to $n$.

**28** [Foundational] • Target: Finding Specific Terms

Find the 4th term in the expansion of $(x+3)^7$.

**a.** $945x^4$ ✓

**b.** $315x^4$

**c.** $2835x^3$

**d.** $105x^5$

For the 4th term, $r=3$. $T_4=\binom{7}{3}x^{7-3}(3)^3=35\cdot x^4\cdot 27=945x^4$.

© ClipSAT — clipsat.org | All Rights Reserved

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** **[Intermediate]** • **Target: Finding Specific Terms**

Find the coefficient of $x^3$ in the expansion of $(2x-3)^6$.

**a.** $-4320$ ✓ **b.** $4320$

**c.** $-1440$ **d.** $2160$

We want $x^3$, so power of $(2x)$ is 3, which means $n-r=3 \Rightarrow r=3$. Term is $\binom{6}{3}(2x)^3(-3)^3 = 20(8x^3)(-27) = -4320x^3$.

**30** **[Intermediate]** • **Target: Term Independent of x and Sum of Coefficients**

Find the term independent of $x$ (the constant term) in the expansion of $\left(x+\frac{2}{x}\right)^6$.

**a.** $160$ ✓ **b.** $80$

**c.** $64$ **d.** $20$

General term: $\binom{6}{r}x^{6-r}(2x^{-1})^r = \binom{6}{r}2^r x^{6-2r}$. Set exponent to 0: $6-2r=0 \Rightarrow r=3$. Constant term $= \binom{6}{3}2^3 = 20 \times 8 = 160$.

**31** **[Intermediate]** • **Target: Term Independent of x and Sum of Coefficients**

What is the sum of all coefficients in the expansion of $(2x-3y)^4$?

**a.** $1$ ✓ **b.** $-1$

**c.** $625$ **d.** $81$

To find the sum of coefficients, substitute $x=1$ and $y=1$: $(2(1)-3(1))^4 = (-1)^4 = 1$.

**32** **[Intermediate]** • **Target: Term Independent of x and Sum of Coefficients**

What is the sum of the elements in row $n$ of Pascal's Triangle (i.e. $\sum_{r=0}^{n}\binom{n}{r}$)?

**a.** $2^n$ ✓ **b.** $n^2$

**c.** $2n$ **d.** $2^{n-1}$

By the Binomial Theorem with $a=1, b=1$: $(1+1)^n = 2^n = \sum_{r=0}^{n}\binom{n}{r}$.

© ClipSAT — clipsat.org | All Rights Reserved

**33** [Intermediate] • Target: Finding Specific Terms

A sequence is defined by $a_1 = 1$ and $a_{n+1} = a_n + 2n + 1$ for $n \geq 1$. Prove by induction that $a_n = n^2$ for all positive integers $n$. Then, consider the series $S_N = \sum_{n=1}^{N} a_n$. Find a closed-form expression for $S_N$ and use it to determine the value of $\sum_{n=1}^{10} (a_n - a_{n-1})$ where $a_0 = 0$.

**a.** **100** ✓

**b.** 110

**c.** 10

**d.** 210

Proof by Induction: Base Case ($n = 1$): $a_1 = 1$. The formula gives $1^2 = 1$. Thus, the base case holds. Inductive Hypothesis: Assume $a_k = k^2$ for some positive integer $k$. Inductive Step: We need to show that $a_{k+1} = (k+1)^2$. From the recurrence relation, $a_{k+1} = a_k + 2k + 1$. Using the inductive hypothesis, $a_{k+1} = k^2 + 2k + 1 = (k+1)^2$. Thus, by induction, $a_n = n^2$ for all positive integers $n$. Closed-form expression for $S_N$: $S_N = \sum_{n=1}^{N} a_n = \sum_{n=1}^{N} n^2$. This is the su.

**34** [Intermediate] • Target: Term Independent of x and Sum of Coefficients

A sequence is defined by the recurrence relation $a_{n+1} = a_n^2 - 2$ for $n \geq 1$. If $a_1 = 2$, find the value of $a_{2023}$.

**a.** **2** ✓

**b.** 0

**c.** -2

**d.** 1

Let's compute the first few terms of the sequence: $a_1 = 2$ $a_2 = a_1^2 - 2 = 2^2 - 2 = 4 - 2 = 2$ $a_3 = a_2^2 - 2 = 2^2 - 2 = 4 - 2 = 2$ It appears that for $a_1 = 2$, all subsequent terms of the sequence are also 2. We can prove this by induction. Base case: For $n = 1$, $a_1 = 2$, which is true. Inductive step: Assume that for some integer $k \geq 1$, $a_k = 2$. We need to show that $a_{k+1} = 2$. From the recurrence relation, $a_{k+1} = a_k^2 - 2$. By the inductive hypothesis, $a_k = 2$. Substituting this into.

**35** [Intermediate] • Target: Binomial coefficients

Find the sum of the first 10 terms of the arithmetic series with $a_1 = -1$ and $d = 3$.

**a.** 128

**b.** 135

**c.** 115

**d.** **125** ✓

$a_{10} = -1 + (10-1)(3) = 26$. $S_{10} = \frac{10}{2}(a_1 + a_n) = \frac{10}{2}(-1 + 26) = 125$.

© ClipSAT — clipsat.org | All Rights Reserved

**36** [Intermediate] • Target: Expansions and specific terms

Find the sum of the first 12 terms of the arithmetic series with $a_1 = -4$ and $d = 3$.

**a.** 138

**b.** 153

**c.** **150** ✓

**d.** 162

$a_{12} = -4 + (12 - 1)(3) = 29$. $S_{12} = \frac{12}{2}(a_1 + a_n) = \frac{12}{2}(-4 + 29) = 150$.

**37** [Intermediate] • Target: More expansions and terms

Find the sum of the first 10 terms of the arithmetic series with $a_1 = -5$ and $d = 5$.

**a.** 185

**b.** **175** ✓

**c.** 180

**d.** 165

$a_{10} = -5 + (10 - 1)(5) = 40$. $S_{10} = \frac{10}{2}(a_1 + a_n) = \frac{10}{2}(-5 + 40) = 175$.

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Sums of binomial coefficients • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Find the sum of the first 10 terms of the arithmetic series with $a_1 = -6$ and $d = 2$.

**a.** 32

**b.** 20

**c.** **30** ✓

**d.** 40

$a_{10} = -6 + (10 - 1)(2) = 12$. $S_{10} = \frac{10}{2}(a_1 + a_n) = \frac{10}{2}(-6 + 12) = 30$.

**39** [Challenging] • Target: Further expansions • Cross-Lesson: 6.02 Polar Equations and Graphs

Find the sum of the first 12 terms of the arithmetic series with $a_1 = -1$ and $d = 1$.

**a.** 66

**b.** 42

**c.** 55

**d.** **54** ✓

$a_{12} = -1 + (12 - 1)(1) = 10$. $S_{12} = \frac{12}{2}(a_1 + a_n) = \frac{12}{2}(-1 + 10) = 54$.

© ClipSAT — clipsat.org | All Rights Reserved

**40** [Challenging] • Target: Deriving Pascal's triangle recursively • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

Find the sum of the first 10 terms of the arithmetic series with $a_1 = -4$ and $d = 2$.

**a.** 40

**b.** **50** ✓

**c.** 60

**d.** 52

$a_{10} = -4 + (10 - 1)(2) = 14$. $S_{10} = \frac{10}{2}(a_1 + a_n) = \frac{10}{2}(-4 + 14) = 50$.

**41** [Challenging] • Target: The general term formula • Cross-Lesson: 7.02 Logarithms and Their Properties

Find the sum of the first 10 terms of the arithmetic series with $a_1 = -1$ and $d = 2$.

**a.** 70

**b.** 90

**c.** **80** ✓

**d.** 82

$a_{10} = -1 + (10 - 1)(2) = 17$. $S_{10} = \frac{10}{2}(a_1 + a_n) = \frac{10}{2}(-1 + 17) = 80$.

**42** [Challenging] • Target: Combinatorial interpretation of binomial coefficients • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

Find the sum of the first 12 terms of the arithmetic series with $a_1 = 0$ and $d = 4$.

**a.** 268

**b.** **264** ✓

**c.** 276

**d.** 252

$a_{12} = 0 + (12 - 1)(4) = 44$. $S_{12} = \frac{12}{2}(a_1 + a_n) = \frac{12}{2}(0 + 44) = 264$.

**43** [Challenging] • Target: Applying the binomial theorem to probability-style counting • Cross-Lesson: 7.04 Applications: Growth and Decay

Find the sum of the first 12 terms of the arithmetic series with $a_1 = 5$ and $d = 2$.

**a.** **192** ✓

**b.** 180

**c.** 204

**d.** 194

$a_{12} = 5 + (12 - 1)(2) = 27$. $S_{12} = \frac{12}{2}(a_1 + a_n) = \frac{12}{2}(5 + 27) = 192$.

© ClipSAT — clipsat.org | All Rights Reserved

**44** [Challenging] • Target: A closing challenge problem • Cross-Lesson: 8.01 Rational Functions and Asymptotes

Find the sum of the first 10 terms of the arithmetic series with $a_1 = 1$ and $d = 1$.

**a.** 56

**b.** **55** ✓

**c.** 45

**d.** 65

$a_{10} = 1 + (10 - 1)(1) = 10$. $S_{10} = \frac{10}{2}(a_1 + a_n) = \frac{10}{2}(1 + 10) = 55$.

**45** [Challenging] • Target: Pascal's Triangle and Combinations • Cross-Lesson: 8.02 Graphing Rational Functions

Find the sum of the first 10 terms of the arithmetic series with $a_1 = -3$ and $d = 1$.

**a.** 16

**b.** **15** ✓

**c.** 5

**d.** 25

$a_{10} = -3 + (10 - 1)(1) = 6$. $S_{10} = \frac{10}{2}(a_1 + a_n) = \frac{10}{2}(-3 + 6) = 15$.

© ClipSAT — clipsat.org | All Rights Reserved